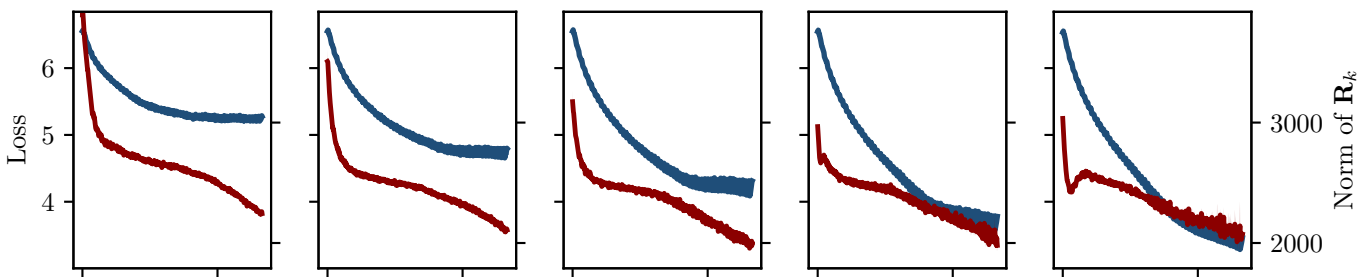
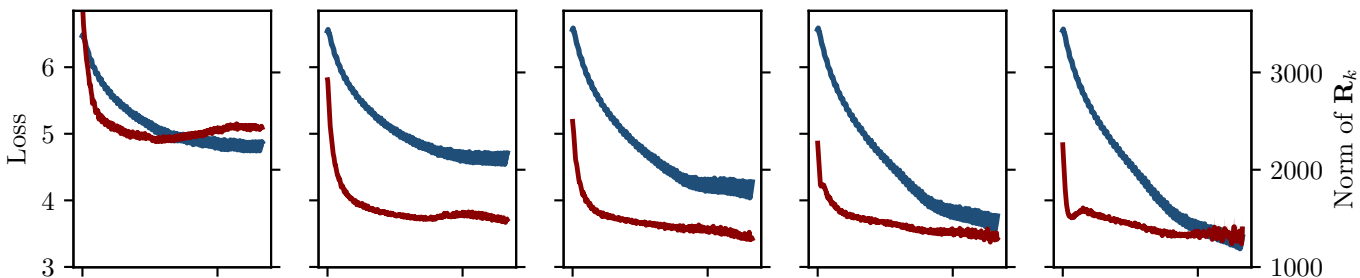


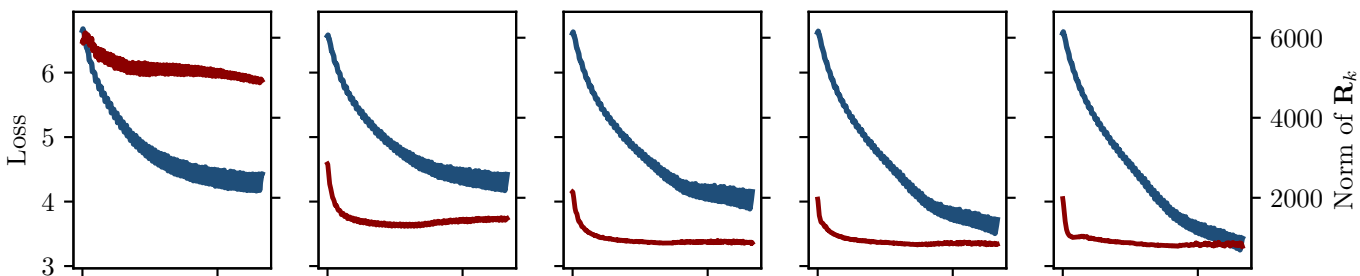
$\beta_1 = 0.900, \beta_2 = 0.900$ $\beta_1 = 0.900, \beta_2 = 0.968$ $\beta_1 = 0.900, \beta_2 = 0.990$ $\beta_1 = 0.900, \beta_2 = 0.9968$ $\beta_1 = 0.900, \beta_2 = 0.999$
 $\omega = 0.4118$ $\omega = 0.447$ $\omega = 0.4937$ $\omega = 0.5079$ $\omega = 0.4759$



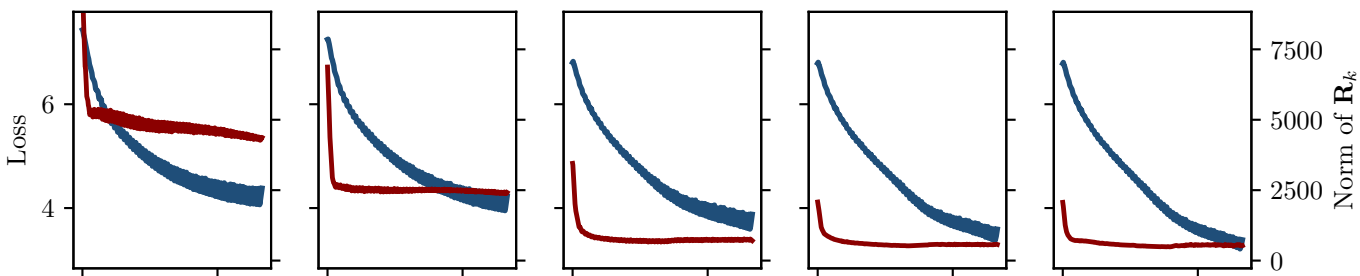
$\beta_1 = 0.968, \beta_2 = 0.900$ $\beta_1 = 0.968, \beta_2 = 0.968$ $\beta_1 = 0.968, \beta_2 = 0.990$ $\beta_1 = 0.968, \beta_2 = 0.9968$ $\beta_1 = 0.968, \beta_2 = 0.999$
 $\omega = 0.2653$ $\omega = 0.3163$ $\omega = 0.2917$ $\omega = 0.2838$ $\omega = 0.2647$



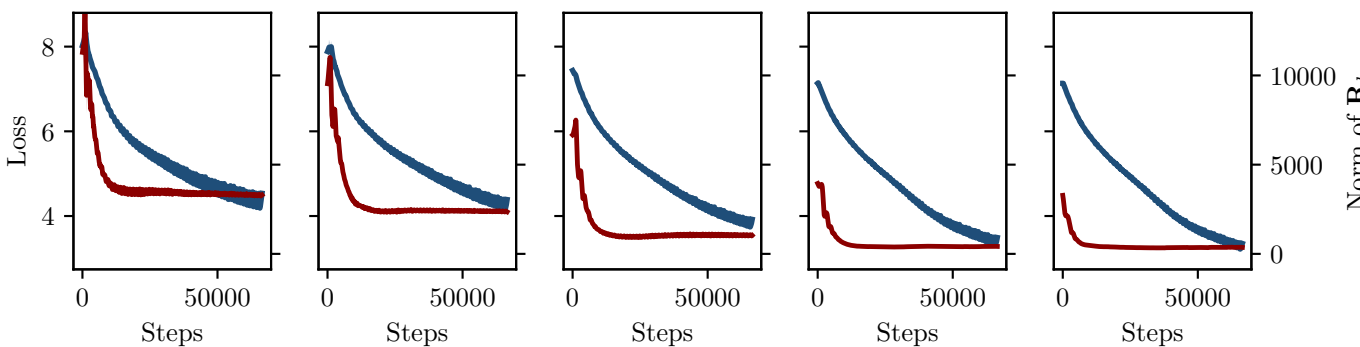
$\beta_1 = 0.990, \beta_2 = 0.900$ $\beta_1 = 0.990, \beta_2 = 0.968$ $\beta_1 = 0.990, \beta_2 = 0.990$ $\beta_1 = 0.990, \beta_2 = 0.9968$ $\beta_1 = 0.990, \beta_2 = 0.999$
 $\omega = 0.6038$ $\omega = 0.1666$ $\omega = 0.1571$ $\omega = 0.161$ $\omega = 0.1489$



$\beta_1 = 0.9968, \beta_2 = 0.900$ $\beta_1 = 0.9968, \beta_2 = 0.968$ $\beta_1 = 0.9968, \beta_2 = 0.990$ $\beta_1 = 0.9968, \beta_2 = 0.9968$ $\beta_1 = 0.9968, \beta_2 = 0.999$
 $\omega = 0.7181$ $\omega = 0.2517$ $\omega = 0.1372$ $\omega = 0.09191$ $\omega = 0.1004$



$\beta_1 = 0.999, \beta_2 = 0.900$ $\beta_1 = 0.999, \beta_2 = 0.968$ $\beta_1 = 0.999, \beta_2 = 0.990$ $\beta_1 = 0.999, \beta_2 = 0.9968$ $\beta_1 = 0.999, \beta_2 = 0.999$
 $\omega = 0.7998$ $\omega = 0.3835$ $\omega = 0.2658$ $\omega = 0.08935$ $\omega = 0.07247$



— Train loss — $\|\mathbf{R}_k\|$